Ambulo - Hiking App

Statement Of Work

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Abstract

Nature hikes are a popular leisure activity worldwide. However, planning them is often complex, requiring information from various sources like maps, weather apps, and recommendation platforms. Consolidating this data and adjusting it to personal preferences can be time-consuming.

Hikers also face unexpected challenges, such as confusing trail splits or sudden weather changes. Limited access to real-time updates may affect safety and enjoyment.

Ambulo addresses these issues with an all-in-one hiking solution. Using advanced technologies, it provides personalized route planning, real-time navigation, and a community platform for sharing experiences.

The app uses algorithms to match routes with user profiles and current conditions. It delivers real-time terrain alerts, weather updates, for a seamless hiking experience.

Ambulo sources its data from internal databases, external APIs, and user input. It is built around three main components:

- **Algorithm Development** – Personalizes route suggestions and equipment advice.
- **SaaS Platform** – Offers maps, alerts, and community reviews across devices.
- **Backend Infrastructure** – Manages data efficiently and securely.

Ambulo's mission is to make hiking more accessible, enjoyable, and safe for everyone.

Introduction

Motivation Hiking is an increasingly popular activity, offering physical, mental, and emotional benefits. Yet, planning a hiking trip remains a fragmented and cumbersome task. It requires collecting and evaluating data from diverse and often inconsistent sources trail maps, weather apps, user forums, and equipment lists making it time-consuming and prone to error. Furthermore, during the hike, real-time updates such as sudden weather shifts, unclear trail splits, or hazards are often unavailable, exposing hikers to risks and frustration.

Problem Statement While several hiking applications exist today, many are limited in scope or specialize in specific regions or features. There is still a need for platforms that offer a more holistic, adaptive, and user-centered hiking experience. Ambulo joins this landscape as an additional solution focused on combining personalized recommendations, real-time updates, and community engagement into a single cohesive platform.

Project Goals and Objectives The primary goal of the Ambulo project is to create a seamless, intelligent, and accessible hiking experience. To achieve this, the system is designed around several key objectives:

- **Personalized Route Selection** – Recommend routes based on user preferences.
- **Comprehensive Route Descriptions** – Include detailed textual and visual information such as photos and GPX files.
- **User Progression System** – Encourage engagement by unlocking new titles as users contribute more.
- **Weather Forecast Integration** – Display accurate 5-day forecasts to help users plan effectively.
- **Email-Based Login** – Enable quick and secure access.
- **Cloud Synchronization** – Ensure that all user data is stored and synchronized across devices (Web & Android).
- **Real-Time Location Features** – Track the user’s location and deliver alerts based on context.
- **Interactive Maps** – Provide dynamic, user-friendly maps with detailed route overlays.

Overview of Our Approach and Contributions Ambulo integrates a cloud-based SaaS infrastructure with advanced algorithm personalization to provide an end-to-end solution for hikers. Our main contributions include:

- A personalized recommendation system for hiking routes
- Real-time alerts and context-sensitive notifications
- Seamless synchronization and offline access via cloud services
- A gamified community model that encourages trail sharing and reviews
- An intuitive, cross-platform mobile application developed with Flutter

By combining intelligent route planning, real-time data, and community input, Ambulo offers a safer, smarter, and more enjoyable hiking experience.

Background and Related Work

Numerous digital tools support hikers today, ranging from navigation-focused platforms to route discovery communities. Notable existing solutions include AllTrails, Komoot, and Amud Anan:

- **AllTrails** offers a robust catalog of trails worldwide, complete with community reviews and GPX downloads. It is especially strong in curated content but locks advanced features like offline maps and real-time alerts behind a paywall.

- **Komoot** combines trip planning with cycling and hiking navigation. It supports personalized planning and turn-by-turn navigation but lacks community-sourced hazard alerts.

- **Amud Anan**, popular in Israel, features community-driven trails with strong local relevance but limited real-time or AI capabilities.

These platforms highlight the market's maturity and user interest but also reveal key limitations that Ambulo aims to improve:

- **Fragmented Functionality** – Features like weather, route suggestions, and community input are often siloed across apps.

- **Lack of Gamification and Contribution Incentives** – User contribution is often voluntary and unstructured, with no progression system.

Ambulo addresses these gaps with:

- **Hybrid Recommendation Algorithms** – Combining content-based and collaborative filtering (Lops et al., 2011; Sarwar et al., 2001).

- **Community-Driven Hazard Reporting** – Enabling real-time input about obstacles and changes in trail conditions.

- **Progression System** – Encouraging user engagement through levels and feature unlocking.

Trail Data Collection Process

To build our trail database, we manually curated high-quality hiking routes over a focused three-week period. We began by exploring popular and trusted online sources to identify trails we were familiar with and considered worthwhile. For each selected trail, we created a custom GPX file to represent the route accurately. We then mapped each trail to match the internal data structure used in our system, recording relevant parameters such as difficulty, length, terrain type, elevation, and more. Once development of the app was complete, all trial data was manually entered into the system. Currently, the app includes nearly 30 curated trails, totaling around 240 kilometers of mapped hiking paths.

System Design / Methods / Approach

Functional Requirements

Ambulo supports a wide range of features aimed at enhancing the user experience:

- **User Authentication and Profiles**: Users sign in securely via Firebase Authentication. Profiles store hiking preferences, history, and contributions.

- **Personalized Route Planning**: Users receive recommended trails based on preferences, difficulty, and location. Advanced filtering includes attributes like distance, terrain, loop type, water sources, season, and accessibility.

- **Comprehensive Route Descriptions**: Each trail includes detailed metadata such as length, elevation gain, time estimate, key landmarks, GPX files, and images.

- **Interactive Maps**: Real-time location tracking, zoomable map views, and toggles for terrain, satellite, and trail conditions are supported.

- **Real-Time Alerts**: The map displays hazards and waypoints.

- **Weather Forecast Integration**: Real-time 5-day forecasts are shown using third-party APIs.

- **Community Features**: Users can share photos, write reviews, and suggest new trails.

- **Navigation Assistance**: GPS-based, turn-by-turn trail navigation is available.

- **Offline Access and Sync**: Local caching ensures offline functionality; cloud sync maintains data consistency across devices.

Non-Functional Requirements

- **Performance**: Capable of handling over 50,000 read/write operations via Firebase.
- **Scalability**: Designed using the SOLID principles to allow easy future migration from Firebase.
- **Security**: Implements secure Firebase Authentication and Firestore rules.
- **Usability**: Offers a clean, responsive UI with intuitive navigation on all device types.
- **Maintainability**: Codebase is modular, documented, and managed via GitHub with continuous integration.
- **Compatibility**: Compatible with Chrome, Firefox, and Edge, supports Android and web.
- **Testing & QA**: Automated tests run via GitHub Actions and local test suites.

System Architecture

Ambulo uses a SaaS-based architecture:

- **Client Interface**: A Flutter-based cross-platform mobile and web app.
- **Cloud Backend**: Business logic, APIs, and data management reside on Firebase Cloud Functions and Firestore.
- **Databases**: Firestore (NoSQL) is used for user data and trail metadata.
- **APIs**: RESTful APIs connect the frontend to the backend and third-party services.

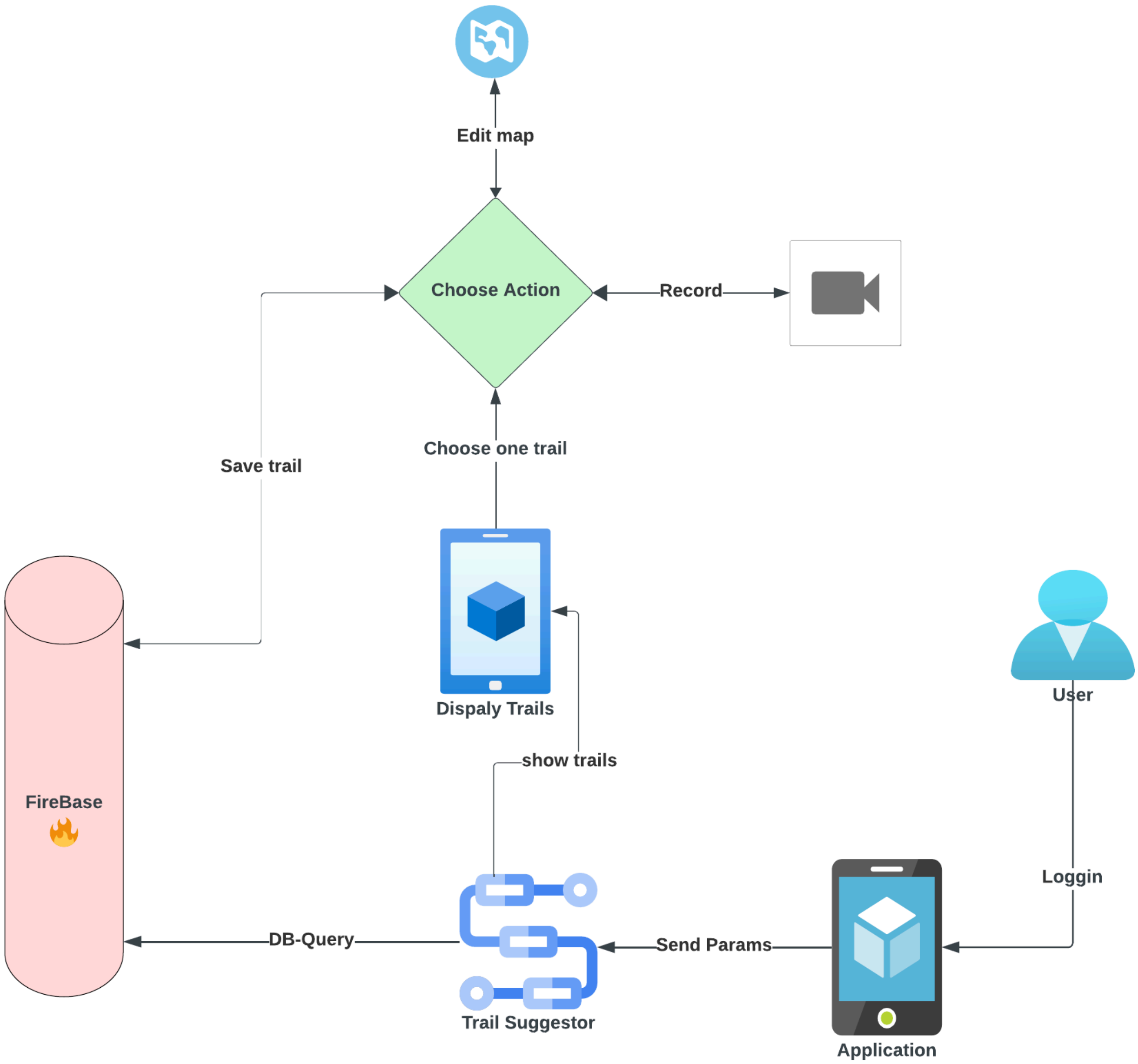
Technology Stack

- **Frontend & Backend**: Flutter (Dart)
- **Database**: Firebase Firestore (NoSQL)
- **Auth**: Firebase Authentication
- **CI/CD**: GitHub + GitHub Actions
- **Weather API**: Integrated via HTTP client packages
- **Mapping**: GPX parsing libraries and map services such as Mapbox or Google Maps

Diagrams

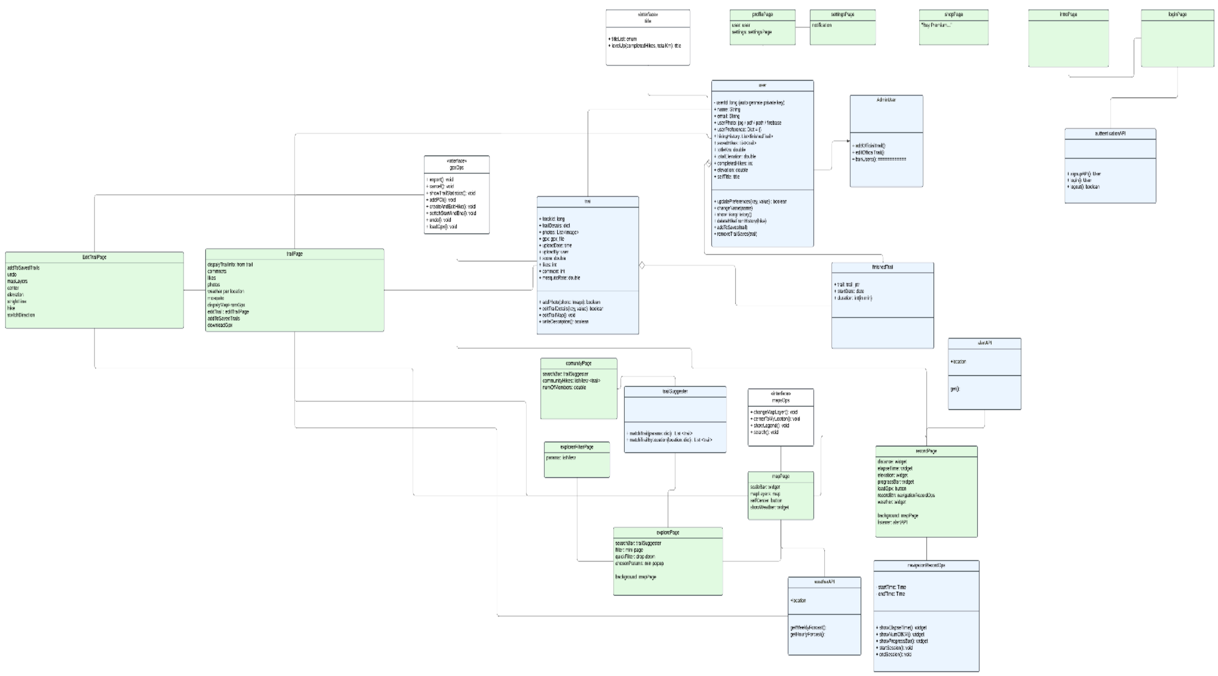
High-level Architecture

[Link to the SVG]



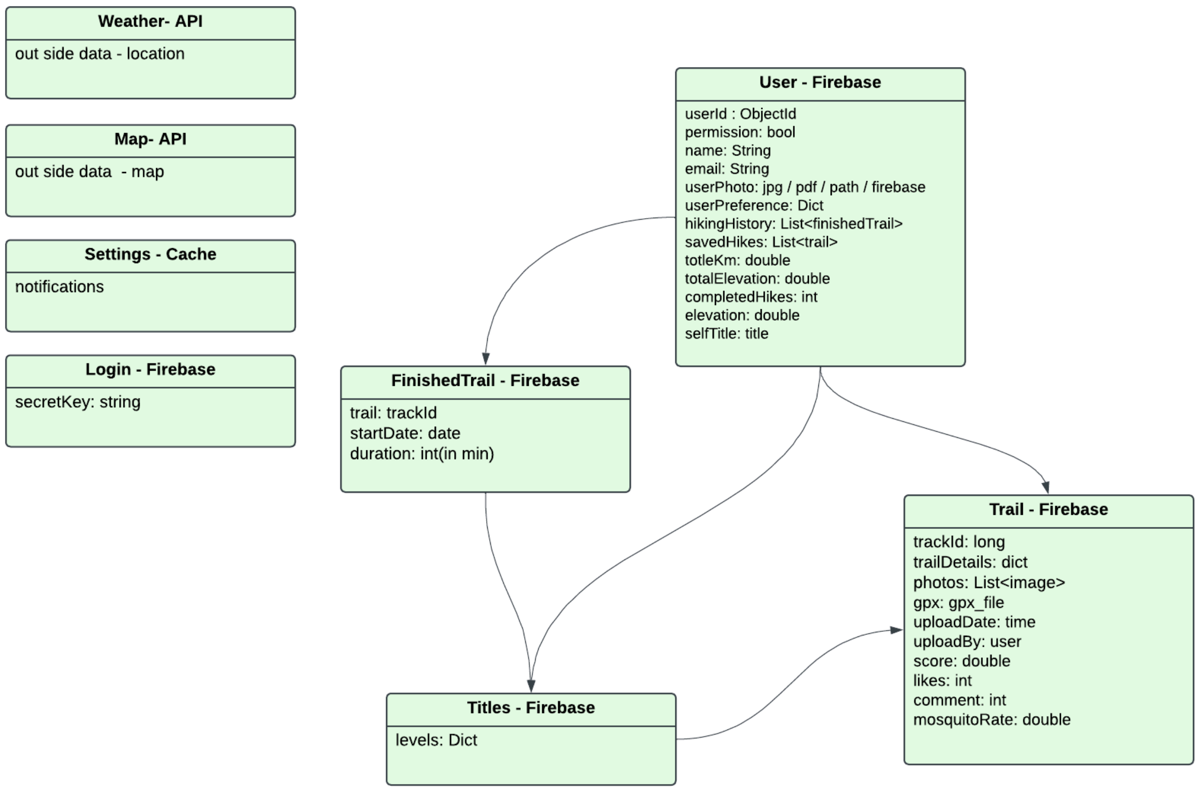
UML - Class

[Link to the SVG]



DataBase Schema

[Link to the SVG]



Methodology

The development process followed an iterative, agile-inspired approach that allowed for frequent testing and feedback. Our dataset was constructed manually and includes nearly 30 curated trails (approximately 240 km). For each trail, we created a custom GPX file, mapped metadata such as length, difficulty, and terrain type, and then uploaded the data manually to the cloud database.

The algorithms used for route recommendation combine collaborative filtering and content-based filtering. We designed a hybrid system that considers user preferences and trail attributes to suggest relevant hikes. All model logic is embedded in the client-facing app.

We used Flutter (Dart) for both frontend and backend logic, Firebase Firestore as our NoSQL cloud database, and Firebase Auth for secure authentication. Map rendering and navigation relied on GPX parsing combined with layers like the Israel Hiking Map, which helped us deliver rich local mapping capabilities.

Our most significant challenges included building a robust and intuitive image upload system, which required careful optimization and integration with cloud storage. Another challenge was building the dataset itself, which involved significant manual work. UI/UX design was also a major focus. We explored references on Mobbin, consulted with developer peers, and tested iteratively with a pilot group of users, including friends and family. Their feedback helped us refine the interface and deliver an app that feels natural to use in real-world hiking conditions.

Key Modules and Components

The Ambulo application is composed of several core modules and data structures:

- **Trail Model**: Represents a hiking trail, including fields like `name`, `description`, `distance`, `difficulty`, `region`, `season`, `surfaceType`, and `gpxPath`. It supports extensibility through `additionalDetails` and media attachment fields.
- **User Model**: Stores user data including `uid`, `email`, `hikingPreferences`, `savedTrails`, and progression level. This model enables personalized recommendations.
- **Recommendation Engine**: A hybrid client-side component that combines collaborative filtering (based on similar users) and content-based filtering (based on trail features) to generate ranked trail suggestions.
- **GPX Parser & Renderer**: Parses `.gpx` files into route segments and visualizes them as polylines on the map using custom renderers with support for terrain overlays and alert markers.
- **Image Management Module**: Handles image upload, compression, and storage via Firebase Cloud Storage. Associates images with specific trails and ensures efficient retrieval.
- **Map Layer Controller**: Manages toggling between map views (satellite, terrain, trail) and overlays such as elevation, weather, and custom trail features.
- **Sync & Offline Cache Layer**: Manages cloud synchronization and caching for trail data and user sessions using Flutter’s local storage and SharedPreferences.

These modules collectively deliver a scalable, modular, and responsive application architecture tailored to the needs of modern hikers.

Discussion

The project highlighted the importance of user-centered design and modular architecture in outdoor apps. While the hybrid recommendation system performs well, it relies on limited initial data. Future iterations can improve personalization, UI responsiveness, and leverage real-time community input more dynamically. Also, we can add in the future the use of AI to provide better results.

Conclusion and Future Work

Ambulo successfully delivers a smart, real-time hiking companion app with personalized routes and live alerts. The system demonstrates strong potential in enhancing hiking experiences through technology.

Planned future work includes:

- Completing the Intro and About pages, including UI screenshots.
- Enhancing the matching algorithm with user contribution scoring.
- Implementing real-time weather alerts.
- Building a full-featured Community page (comments, reviews, suggestions).
- Adding socket-based chat, shared checklists, and group trail planning.
- Supporting offline maps for remote areas.
- Improving database performance through a multi-collection approach.

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